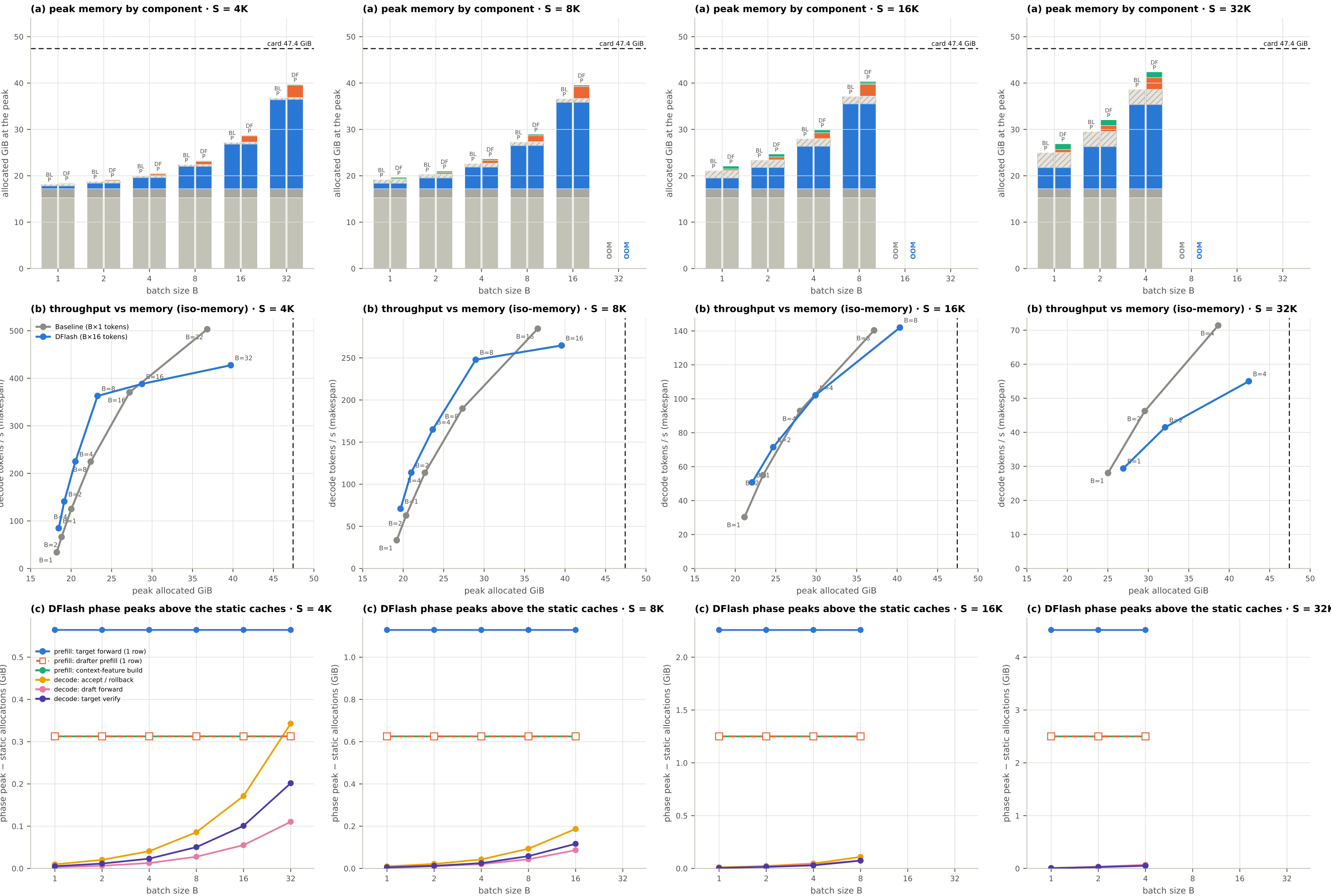


Qwen3-8B — capacity: what sets the peak, and what the drafter adds

target weights target KV (static, all rows) activation transient tapped target hidden (1 row) (a): left bar baseline, right bar DFlash; P / D = peak in prefill / decode
draft weights (loaded in both modes) GDN state (all rows) draft KV (static, all rows) context feature



Components are read at the one instant the run peaked. Prefill is per row (one request at a time) into caches allocated for all B rows at setup, so the peak is the static caches plus ONE row's transient; every OOM happened at 'setup: allocate caches'. Both modes load the drafter's weights (the same process runs both), so a baseline-only process would sit that much lower. (c) is the peak inside each phase minus what setup allocated: the drafter's own prefill never exceeds the target's, because its context tokens only pass through fc and the K/V projections, not through its attention queries or MLP.